

Jinhee Won

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EDUCATION

Massachusetts Institute of Technology (MIT)

M.Eng. in Artificial Intelligence; GPA: 5.0/5.0

Cambridge, MA

Sep 2025 – Expected Dec 2026

- Research advised by Professors Munther Dahleh and Mardavij Roozbehani
- Teaching Assistant: Reinforcement Learning (6.7920) and Advances in Computer Vision (6.8300)

B.S. in Computer Science and Artificial Intelligence; GPA: 4.8/5.0

Sep 2022 – Dec 2025

- Relevant Coursework: Inference and Information (6.7800), Bayesian Inference (6.7830), Time Series Analysis (6.s899), Multiagent Learning (6.s890), Design and Analysis of Algorithms (6.1220), Software Construction (6.1020)

EXPERIENCE

MIT Institute for Data, Systems, and Society

Graduate Researcher

Cambridge, MA

Feb 2025 – Present

- Developed VAE-based synthetic-data and transfer methods to improve downstream performance in low-data classification and reinforcement-learning settings.
- Developed low-rank reinforcement-learning methods applying matrix-estimation techniques to Q-learning, DQN, and PPO.
- Implemented DPO and reward-model training pipelines for LLM post-training, scaling experiments across an HPC cluster.

SERC Scholars Program, MIT

Researcher

Cambridge, MA

Sept 2025 – May 2026

- First-authored *Native-Speaker Personas Are Not Enough: Language Pathways Shape LLM Interpersonal Advice*, accepted to EMNLP 2026 Main Conference.
- Led a 230,000+ response study comparing native-language and native-speaker persona prompting across 600 advice questions, 12 target languages, and 8 open-weight and proprietary LLMs.
- Developed a four-layer linguistic and behavioral evaluation pipeline, finding that persona prompting adds surface-level social cues while reducing concrete, actionable guidance and shifting recommendations toward confrontation.

IMC Trading

Quantitative Trading Intern

Chicago, IL

Jun 2025 – Aug 2025

- Developed a trade-toxicity segmentation pipeline by evaluating XGBoost, k-nearest neighbors, and clustering methods over order-timing, BBO-status, and other market-state features.
- Improved separation in post-trade expected value between predicted groups by 32% relative to the baseline, producing more discriminative toxicity categories.
- Investigated auction and SPX/ES market data to identify and evaluate potential trading signals.

MIT SuperUROP

Undergraduate Researcher

Cambridge, MA

Sep 2024 – May 2025

- Improved surgical-site infection prediction ROC-AUC by 8 percentage points through multimodal feature engineering and systematic model selection.
- Developed models combining RGB and thermal images using data from approximately 700 patients.
- Benchmarked alternative model architectures, transfer-learning strategies, augmentation methods, and class-imbalance techniques under limited clinical data.

Samsara

Firmware Intern

San Francisco, CA

May 2024 – Aug 2024

- Built an ML anomaly-detection system, detecting issues up to nine days earlier than manual review for firmware metrics.
- Improved structured logging for Vehicle Gateway and Camera Manager systems, allowing for consistent collection of on-device debugging information.
- Investigated embedded-device behavior and production telemetry using Python, SQL, Go, and SSH.

SELECTED PROJECTS

LLM-Generated Reward Functions for Reinforcement Learning

- Built a framework for generating and iteratively refining executable reward functions for PPO agents using an LLM and environment feedback.
- Evaluated generated rewards across robotics environments, measuring task success, sample efficiency, and reward hacking against manually designed baselines.

Spatially Conditional Diffusion Models

- Implemented iterative denoising, classifier-free guidance, and time- and class-conditioned U-Nets in PyTorch.
- Extended the denoising process with spatial light conditioning, injecting a parameterized light kernel at each timestep to control illumination in generated images.

SKILLS

- **Languages and Tools:** Python, C++, Go, SQL, Git, Linux, Docker, Datadog, Kubernetes
- **Areas:** reinforcement learning, diffusion models, variational autoencoders, Bayesian inference, time-series analysis, VLAs
- **Interests:** Digital art, photography, bridge, ceramics